NATALIE NEAMTU

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EDUCATION

CARNEGIE MELLON UNIVERSITY — **PITTSBURGH, PA**
DEPARTMENT OF COMPUTER SCIENCE
PhD, Computer Science — *AUGUST 2025 – PRESENT*
Expected graduation: 2030

UNIVERSITY OF WASHINGTON — **SEATTLE, WA**
PAUL G. ALLEN SCHOOL OF COMPUTER SCIENCE & ENGINEERING
PhD, Computer Science (unfinished) — *SEPTEMBER 2024 – AUGUST 2025*

CORNELL UNIVERSITY, COLLEGE OF ARTS & SCIENCES — **ITHACA, NY**
Bachelor of Arts, Computer Science (summa cum laude) & Mathematics — *SEPTEMBER 2018 – MAY 2021*
GPA: 4.21/4.3

RUTGERS UNIVERSITY HONORS COLLEGE — **NEW BRUNSWICK, NJ**
GPA: 4.0/4.0 — *SEPTEMBER 2017 – MAY 2018*

PUBLICATIONS & PREPRINTS

Yang, C., **Neamtu, N.**, Hawblitzel, C., Lorch, J.R. and Lu, S. VeruSAGE: A Study of Agent-Based Verification for Rust Systems. In *Advances in Neural Information Processing Systems 40, Datasets and Benchmarks Track (2026)*. arXiv.

Agarwal, A., **Neamtu, N.**, Aggarwal, P., Kim, S., Limperg, J., Flamant, C., Shimizu, K., Parno, B. and Welleck, S.. Verus-SpecGym: An Agentic Environment for Evaluating Specification Autoformalization. Preprint (2026). arXiv.

Neamtu, N, Ni, H., and Van Renesse, R.. Trees and Turtles: Modular Abstractions for State Machine Replication Protocols. In *Proceedings of the 10th Workshop on Principles and Practice of Consistency for Distributed Data (PaPoC '23)*. arXiv.

TALKS

Towards Provably Correct, Real-World Code Generation. With Pranjal Aggarwal. *Jane Street Research Symposium*, August 2026. Slides.

Supporting Weak Memory in Creusot and Verus. *Rust Verification Workshop*, April 2026. Slides.

INDUSTRY EXPERIENCE

MICROSOFT RESEARCH – SYSTEMS RESEARCH GROUP — **REDMOND, WA**
RESEARCH INTERN — *MAY 2025 – AUGUST 2025*

- Mentors: Chris Hawblitzel, Shan Lu, Jay Lorch.
- Built an agentic system to fully automate proof completion for real-world Verus systems verification projects.

AWARDS

- **NSF CSGrad4US Fellowship** (2023). Selected from early-career individuals with the demonstrated potential to be high-achieving CISE researchers and innovators and to broaden participation among groups underrepresented in CISE disciplines.
- **Cornell University Merrill Presidential Scholar** (2021). Selected from top 1% of graduating class based on academic scholarship, demonstrated intellectual drive, and leadership abilities.
- **Alan S. Marx Memorial Prize for Excellence in Teaching** (2021). Awarded annually to 2 graduating seniors by the Cornell University computer science department.
- **Outstanding TA Award** (2020). Received for service as TA (CS 3110, Cornell University).
- **Frank and Rosa Rhodes Scholarship** (2019). Given to one outstanding scholar annually from each of the seven undergraduate colleges at Cornell University.

TEACHING & OUTREACH

TEACHING ASSISTANT (CORNELL UNIVERSITY) *SEPTEMBER 2019 – MAY 2021*

- Courses: CS 3110 (Functional Programming), CS 4160 (Formal Verification), CS 5432 (Advanced System Security).
- Hold weekly recitation sessions and office hours, grade coding and written assignments, write homework and exam questions, assist in creating course materials.
- **Lead TA for Lab Development** (CS 3110, Fall 2020): Led team of 10 course staff members to develop new weekly lab assignments to be used in current and future iterations of the course.
- Tested assignments and edited course materials for new course (CS 5432, Spring 2021).

GENERAL OUTREACH VOLUNTEER (WICC) *SEPTEMBER 2018 – MAY 2019*

- Volunteered for outreach events and programming contests for high school students at Cornell.

MEMBER (RUTGERS DOUGLASS-DIMACS COMPUTING CORPS) *SEPTEMBER 2017 – MAY 2018*

- Visited local middle schools and hosted field trips young women interested in computing.

PROFESSIONAL EXPERIENCE

MICROSOFT (BING) **REDMOND, WA**
SOFTWARE ENGINEER *AUGUST 2021 – SEPTEMBER 2024*

- Data quality and ingestion for Bing's Knowledge Graph.

MICROSOFT (BING) **REDMOND, WA**
SOFTWARE ENGINEER INTERN *JUNE 2020 – AUGUST 2020*

MICROSOFT (OS DATA INTELLIGENCE) **REDMOND, WA**
EXPLORER INTERN *JUNE 2019 – AUGUST 2019*

BANK OF AMERICA MERRILL LYNCH **NEW YORK, NY**
GLOBAL TECHNOLOGY FRESHMAN SUMMER ANALYST INTERN *JUNE 2018 – AUGUST 2018*